ECE250: Signals & Systems Quiz-7

Date: 29th October, 2025 **Total Marks:** 10 **Duration:** 15 Minutes

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[CO1 CO2 CO4] **[Q1]**: A causal and stable LTI system S has the frequency response

$$H(j\omega) = \frac{j\omega + 4}{6 - \omega^2 + 5j\omega}$$

(a) Determine a differential equation relating the input $x(t)$ and output $y(t)$ of S. **[3 Marks]**

(b) Determine the impulse response $h(t)$ of S. **[3 Marks]**

(c) What is the output of S when the input is **[4 Marks]**

$$x(t) = e^{-4t}u(t) - t^2e^{-4t}u(t)?$$